In [54]:
```
data={
    'Area':[500,600,700,800,900,1000,1200,1400,1600,1800],
    'Bedrooms':[1,1,2,2,2,3,3,3,4,4],
    'Age':[20,18,15,12,10,8,7,5,3,2],
    'Price':[30,35,40,45,50,58,65,75,85,90]
}
```

In [56]:
```
import pandas as pd
df=pd.DataFrame(data)
```

In [58]:
```
df
```

Out[58]:

	Area	Bedrooms	Age	Price
0	500	1	20	30
1	600	1	18	35
2	700	2	15	40
3	800	2	12	45
4	900	2	10	50
5	1000	3	8	58
6	1200	3	7	65
7	1400	3	5	75
8	1600	4	3	85
9	1800	4	2	90

X and Y

In [61]:
```
Y=df[['Price']]
X=df.drop(labels=['Price'],axis=1)
```

In [84]:
```
from sklearn.model_selection import train_test_split
xtrain,xtest,ytrain,ytest=train_test_split(X,Y,test_size=0.2,random_state=21)
```

Create Model

In [87]:
```
from sklearn.tree import DecisionTreeRegressor
model=DecisionTreeRegressor(max_depth=4,random_state=42)
```

Train Model

In [90]:
```
model.fit(xtrain,ytrain)
```

Out[90]:
```
DecisionTreeRegressor(max_depth=4, random_state=42)
```

In [92]:
```
y_pred=model.predict(xtest)
```

In [94]:
```
y_pred
```

Out[94]:
```
array([30., 45.])
```

In [96]:
```
ytest
```

Out[96]:

	Price
1	35
2	40

Model Evalution

In [99]:
```
from sklearn.metrics import mean_absolute_error
mae=mean_absolute_error(ytest,y_pred)
print("MAE: ",mae)
```

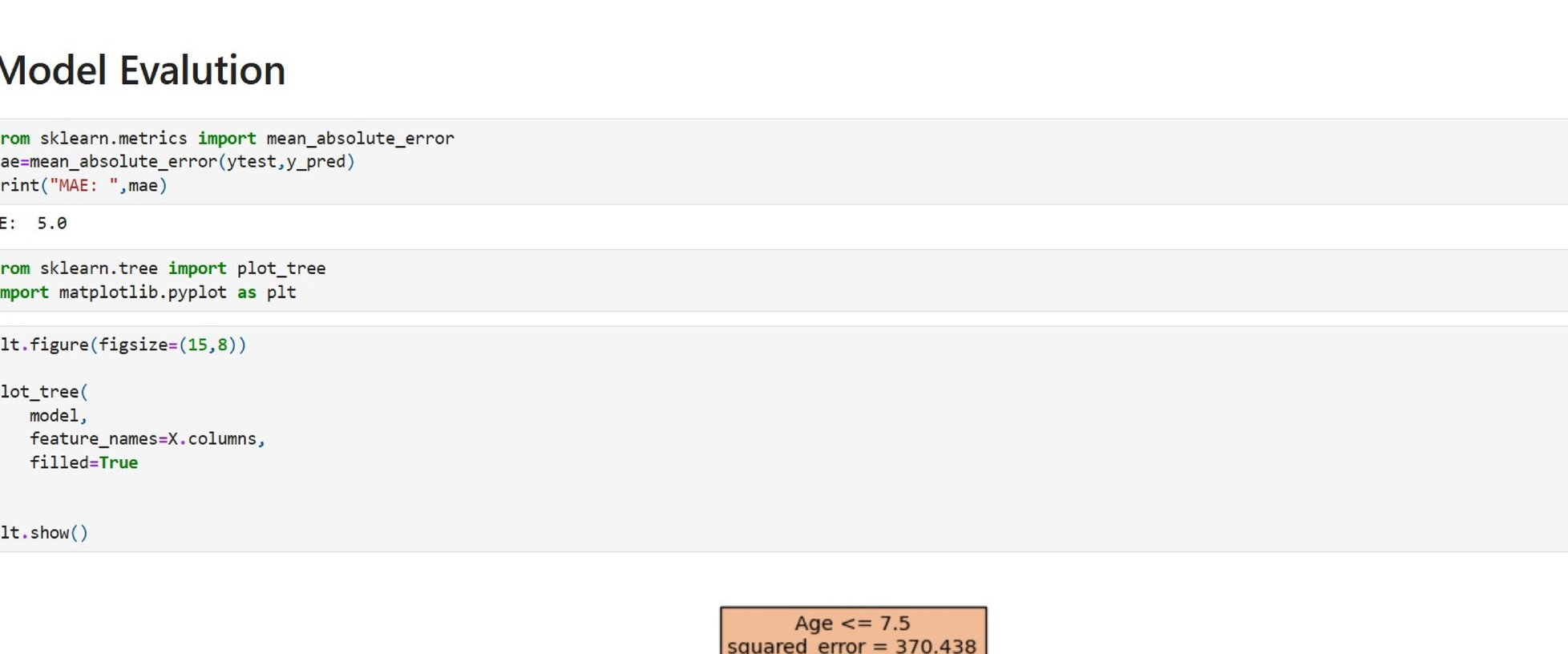
```
MAE:  5.0
```

In [101…
```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt
```

In [108…
```
plt.figure(figsize=(15,8))

plot_tree(
    model,
    feature_names=X.columns,
    filled=True
)

plt.show()
```



In [ ]:

In [ ]: